```
+-------------------------------------------------------------------+
| GPS DENIAL, ELECTRONIC JAMMING & DEAD RECKONING EXECUTION DIRECTIVE |
| FLEET: ARDUPILOT EKF3 UNIVERSAL                  AVIONICS OPERATIONAL SOP |
+-------------------------------------------------------------------+
```

PART I: AVIONICS PARAMETER TECHNICAL REFERENCE

1. GNSS HARDWARE CUTOFF & RF JAMMING SIMULATION

SIM_GPS1_ENABLE

[HARDWARE TOGGLE]

DEFAULT: 1 (ACTIVE) **TARGET:** 0 (SEVERED / DENIED)

LSK 1L COMMAND: param set SIM_GPS1_ENABLE 0

Simulates physical antenna severance or receiver power cutoff. SITL instantly sets fix_type = 0 (NO_FIX) and satellites_visible = 0. EKF3 immediately declares satellite aiding invalid and transitions directly to smooth Dead Reckoning without coordinate jitter.

SIM_GPS1_JAM

[ELECTRONIC WARFARE INJECTION]

DEFAULT: 0 (NORMAL RF) **TARGET:** 1 (RF JAMMED)

LSK 2L COMMAND: param set SIM_GPS1_JAM 1

Injects broadband RF interference on GPS L1/L2 carrier frequencies (C/N0 < 28 dB-Hz), inflating pseudorange noise. **CAUTION:** SITL maintains a degraded 3D fix with jumping coordinates; requires automated gating configuration (Section 2) to eliminate aircraft wandering.

2. INNOVATION GATING & ATTITUDE HORIZON PROTECTION

EK3_GLITCH_RAD

[INNOVATION STEP GATE THRESHOLD]

DEFAULT: 25.0 m **RECOMMENDED:** 15.0 to 20.0 m

MCDU COMMAND: param set EK3_GLITCH_RAD 15

Defines maximum allowable horizontal step jump between consecutive GPS updates before EKF3 rejects the measurement as anomalous. Rejects abrupt adversarial spoofing coordinates and extreme jamming spikes.

AHRS_OPTIONS

[DCM FALLBACK INHIBIT BITMASK]

DEFAULT: 0 (FALLBACK ALLOWED) **ANTI-JAM TARGET:** 3 (DCM LOCKOUT)

MCDU COMMAND: param set AHRS_OPTIONS 3

Critical anti-hunting parameter. Disables automatic fallback from EKF3 to legacy DCM (Bit 0: FW, Bit 1: VTOL). Under RF jamming, DCM bypasses EKF dead reckoning and feeds raw jumping GPS directly to navigation, causing wild wandering.

AHRS_GPS_USE

[ATTITUDE FILTER ISOLATION]

DEFAULT: 1 (ENABLED) **HOSTILE RF TARGET:** 0 (ISOLATE AHRS)

MCDU COMMAND: param set AHRS_GPS_USE 0

Disconnects GPS ground velocity from attitude centrifugal turn compensation (a_cent = omega * V_gps). Prevents erratic GPS velocity spikes from corrupting the artificial horizon and inducing violent wing-rocking.

```
+-------------------------------------------------------------------------+
| SENSOR FALLBACK CASCADE & AERODYNAMIC DRAG WIND ESTIMATION |
| SUBSYSTEM: EKF3 INERTIAL CORE                                AVIONICS OPERATIONAL SOP |
+-------------------------------------------------------------------------+
```

3. DEAD RECKONING & AERODYNAMIC WIND ESTIMATION

EK3_DRAG_BCOEF_X / EK3_DRAG_BCOEF_Y [DRAG BALLISTICS]

DEFAULT: 0.0 (DISABLED) **AIRFRAME TARGET:** 25.0 to 35.0 (kg/m^2)

LSK 5R COMMANDS: `param set EK3_DRAG_BCOEF_X 25` | `param set EK3_DRAG_BCOEF_Y 25`

Calibrates longitudinal and lateral airframe ballistic drag. In GPS-denied flight, EKF3 combines pitot true airspeed (V_{tas}) with accelerometer drag deceleration to estimate horizontal wind ($\vec{W}$), synthesizing ground track: $\vec{V}_{\text{ground}} = \vec{V}_{\text{air}} + \vec{W}$.

EK3_OPTIONS [JAMMING EXPECTED EKF BEHAVIOR]

DEFAULT: 0 (STANDARD EKF) **ANTI-JAM TARGET:** 1 (JAMMING EXPECTED)

LSK 4L COMMAND: `param set EK3_OPTIONS 1`

Bit 0 (JammingExpected). In EKF3, prevents automatic position reset to degraded GPS upon timeout. Sets waitingForGpsChecks = true, isolating the filter in pure Dead Reckoning until all pre-flight quality checks pass continuously.

4. OPTICAL FLOW SIMULATION & HOVER VELOCITY FUSION

SIM_FLOW_ENABLE / EK3_FLOW_USE / EK3_SRC1_VELXY [FLOW FUSION]

DEFAULT: 0 / 1 / 3 **EXECUTION TARGET:** 1 / 1 / 5

MCDU COMMANDS: `param set SIM_FLOW_ENABLE 1` | `param set EK3_SRC1_VELXY 5`

Activates simulated downward optical flow camera and directs EKF3 to ingest pixel displacement rates as primary horizontal velocity (V_x, V_y), decoupling hover navigation from satellites.

5. ENVIRONMENTAL WIND & DRYDEN TURBULENCE INJECTION

SIM_WIND_SPD / SIM_WIND_DIR / SIM_WIND_TURB [METEOROLOGY]

RANGE: SPD: 0--30 m/s | DIR: 0--359° | TURB: 0.0--1.0

MCDU COMMANDS: `param set SIM_WIND_SPD 12` | `param set SIM_WIND_DIR 270`

Injects steady crosswinds and Dryden atmospheric gusts into SITL to validate crab angle computation and EKF3 wind triangle estimation under realistic turbulence.

```
+-----------------------------------------------------------------------+
| TACTICAL FLIGHT SCENARIOS 1 & 2 EXECUTION PROTOCOLS |
| TOTAL GNSS HARDWARE CUTOFF & ELECTRONIC WARFARE JAMMING COUNTER |
+-----------------------------------------------------------------------+
```

SCENARIO 1: TOTAL GNSS HARDWARE CUTOFF [CLEAN DEAD RECKONING]

OBJECTIVE: Evaluate pure aerodynamic dead reckoning upon complete loss of primary satellite reception (antenna severance / power loss). Verify track hold without wander.

PREREQUISITES: Airspeed valid; cruise in AUTO at 50--100 m AGL; drag set: EK3_DRAG_BCOEF_X/Y = 25.

LSK 1L TRIGGER: `param set SIM_GPS1_ENABLE 0`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- fix_type drops instantly to 0 (NO_FIX), satellites_visible = 0.
- EKF3 cleanly decouples GPS horizontal fusion; transitions immediately to Dead Reckoning.
- Aircraft holds track via airspeed and compass heading; zero wandering or wing-rock.

RECOVERY: `param set SIM_GPS1_ENABLE 1` ($\rightarrow$ GPS lock restored in 3--5 s).

SCENARIO 2: EW RF JAMMING COUNTERMEASURE [ANTI-HUNT PROTOCOL]

OBJECTIVE: Counter broadband RF jamming. Prevent coordinate chasing (''here-and-there'' hunting) by locking out legacy DCM fallback and locking EKF3 in aerodynamic Dead Reckoning.

ROOT CAUSE OF WANDERING: (1) Default AHRS_OPTIONS = 0 falls back to DCM when EKF rejects GPS but status >= 3D; DCM has *no* dead reckoning and feeds raw jumping GPS to navigation. (2) Without EK3_OPTIONS = 1, EKF resets position to bad GPS on timeout. (3) Dead reckoning requires a converged wind vector: $\vec{V}_{\text{gnd}} = \vec{V}_{\text{tas}} + \vec{W}$.

STEP 1 --- PRE-FLIGHT ANTI-JAM SETUP (LSK 4L, 4R, 5L, 5R):

```
param set AHRS_OPTIONS 3    [Inhibits DCM fallback; stays in EKF3]
param set EK3_OPTIONS 1     [JammingExpected; blocks bad position reset]
param set AHRS_GPS_USE 0    [Severs GPS velocity from attitude horizon]
param set EK3_DRAG_BCOEF_X 25    [Ballistic drag for wind vector solution]
```

STEP 2 --- MANDATORY WIND-LEARNING ORBIT (PRIOR TO JAMMING):

Fly AUTO or LOITER turns for **20 seconds** with valid GPS to converge wind states ($\vec{W}$).

STEP 3 --- INJECT RF JAMMING (LSK 2L TRIGGER): `param set SIM_GPS1_JAM 1`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- SITL injects noise; EKF3 detects innovation errors and isolates GPS inputs.
- AHRS_OPTIONS 3 blocks DCM fallback; navigation stays strictly in EKF3.
- EK3_OPTIONS 1 holds Dead Reckoning without resetting position to noise.
- Aircraft tracks waypoints accurately; wandering and hunting are eliminated.

RECOVERY: `param set SIM_GPS1_JAM 0` $\rightarrow$ `param set AHRS_GPS_USE 1`

```
+-----------------------------------------------------------------------+
| TACTICAL FLIGHT SCENARIOS 3 & 4 EXECUTION PROTOCOLS |
| SPOOFING GATING DEFENSE & OPTICAL FLOW HOVER FALLBACK |
+-----------------------------------------------------------------------+
```

SCENARIO 3: ADVERSARIAL GPS SPOOFING STEP-JUMP DEFENSE

OBJECTIVE: Validate EKF3 innovation gating defense against hostile terrestrial spoofers attempting to hijack the aircraft trajectory via false pseudoranges or sudden coordinate offset steps.

STEP 1 --- GATING DEFENSIVE PRE-CONFIGURATION:

`param set EK3_GLITCH_RAD 15` | `param set AHRS_GPS_USE 0`

STEP 2 --- INJECT COORDINATE STEP OFFSET (LSK 3L TRIGGER):

`param set SIM_GPS1_GLT_X 60` | `param set SIM_GPS1_GLT_Y 60`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- Spoofed coordinate jump (≈ 85 m) exceeds the EK3_GLITCH_RAD 15 m gate.
- EKF3 flags GPS_Glitch: pos step in MAVLink STATUSTEXT; false vector is rejected.
- Aircraft continues holding planned track on internal dead reckoning, completely ignoring spoofer pull.

RECOVERY COMMANDS:

`param set SIM_GPS1_GLT_X 0` | `param set SIM_GPS1_GLT_Y 0` | `param set AHRS_GPS_USE 1`

SCENARIO 4: VTOL / LOW-ALTITUDE HOVER OPTICAL FLOW FALLBACK

OBJECTIVE: Ensure stationary hover and station-keeping capability for VTOL platforms or quadplanes when GPS is denied in zero-airspeed hover flight.

STEP 1 --- SENSOR PRE-CONFIGURATION:

`param set SIM_FLOW_ENABLE 1` | `param set EK3_FLOW_USE 1` | `param set EK3_SRC1_VELXY 5`

STEP 2 --- FLIGHT REGIME ESTABLISHMENT:

Establish aircraft in steady hover (QLOITER or QHOVER mode) at 5--15 m AGL over textured ground.

STEP 3 --- SEVER GPS IN HOVER (LSK 1L TRIGGER):

`param set SIM_GPS1_ENABLE 0`

TELEMETRY DYNAMICS & FLIGHT RESPONSE:

- MAVLink OPTICAL_FLOW packets stream angular ground pixel displacement rates (rad/s).
- EKF3 seamlessly substitutes satellite horizontal velocity with integrated optical flow ground velocity.
- Aircraft holds stationary hover coordinates without drift, even with zero forward airspeed over pitot tube.

RECOVERY COMMANDS:

`param set SIM_GPS1_ENABLE 1` | `param set EK3_SRC1_VELXY 3`

```
+-----------------------------------------------------------------------+
| MASTER FLIGHT TEST MATRIX & COCKPIT QUICK-KEY REFERENCE |
| OPERATIONAL FLIGHT TEST CHECKLIST                                     PAGE: 5 OF 5 |
+-----------------------------------------------------------------------+
```

SCENARIO 5: HIGH-CROSSWIND TURBULENCE DEAD RECKONING

OBJECTIVE: Validate EKF3 wind triangle estimation ($\vec{V}_{\text{ground}} = \vec{V}_{\text{tas}} + \vec{W}_{\text{wind}}$) and crab compensation under crosswinds and turbulence during GPS denial.

STEP 1 --- INJECT CROSSWIND METEOROLOGY:

`param set SIM_WIND_SPD 12` | `param set SIM_WIND_DIR 270` | `param set SIM_WIND_TURB 0.25`

STEP 2 --- SEVER GPS UNDER CROSSWIND:

`param set SIM_GPS1_ENABLE 0`

Response: Aircraft computes crab angle ($\beta \approx 35°$), holding ground track true.

RECOVERY COMMANDS:

`param set SIM_GPS1_ENABLE 1` | `param set SIM_WIND_SPD 0`

PART III: MASTER FLIGHT TEST MATRIX

SCENARIO	PRE-SETUP	TRIGGER	KEY TELEMETRY	EVALUATION
1. CLEAN LOSS	Airspeed valid, BCOEF = 25	`param set SIM_GPS1_ENABLE 0`	GPS.fix = 0 GPS.sats = 0	Smooth transition to Dead Reckoning; zero wandering or wing-rock.
2. JAM COUNTER	AHRS_OPT 3 EK3_OPT 1 AHRS_GPS 0 20s Wind Orbit	`param set SIM_GPS1_JAM 1`	DCM lockout waitingChecks 1 DR active	DCM lockout active; EKF holds Dead Reckoning without coordinate reset; zero wandering.
3. SPOOF DEFENSE	GLITCH_RAD 15 AHRS_GPS 0	`param set SIM_GPS1_GLT_X 60`	GPS_Glitch in STATUSTEXT	Measurement rejected at gate; internal dead reckoning maintained.
4. FLOW HOVER	FLOW_ENABLE 1 SRC1_VELXY 5	`param set SIM_GPS1_ENABLE 0`	OPTICAL_FLOW rate > 20 Hz	Zero positional drift in low-altitude hover; velocity aided by camera.
5. HIGH-WIND DR	WIND_SPD 12 WIND_DIR 270	`param set SIM_GPS1_ENABLE 0`	Crab angle established; track true	Correct crosswind crab computed from drag model; path held in gusts.

COCKPIT MCDU / LSK QUICK-KEY MAP REFERENCE:

```
< LSK 1L: Clean GPS Cutoff (SIM_GPS1_ENABLE 0)          LSK 1R >: RECOVER GPS (ENABLE 1)
< LSK 2L: RF Jamming Noise (SIM_GPS1_JAM 1)             LSK 2R >: CLEAR JAMMING (JAM 0)
< LSK 3L: Spoofing Offset (SIM_GPS1_GLT_X/Y)            LSK 3R >: RESET GLITCH OFFSETS
< LSK 4L: Jamming Expect (EK3_OPTIONS 1)          LSK 4R >: DCM Lockout (AHRS_OPTIONS 3)
< LSK 5L: Isolate AHRS (AHRS_GPS_USE 0)             LSK 5R >: Drag Coeff (BCOEF_X 25)
< LSK 6L: OPEN THIS SOP MANUAL                           LSK 6R >: RETURN TO NAV POS
```